

## BEDFORD BASIN, NS, Canada

WMO: 713290

Lat: 44.710N

Lon: 63.630W

Elev: 4

StdP: 101.28

Time Zone: -4.00 (NAA)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.7      | -11.7 | -17.5                         | 0.8 | -11.7 | -15.9 | 0.9 | -9.6 | 15.8                  | 8.8  | 13.8 | 2.8  | 4.4                   | 320  | 0.597 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 7.6                    | 26.9            | 19.5 | 25.3 | 18.9 | 23.9 | 18.4 | 21.5                | 24.5 | 20.6 | 23.3 | 19.7 | 22.2 | 4.1                  | 300  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.6                            | 15.3 | 22.3 | 19.7 | 14.4 | 21.7 | 18.8 | 13.6 | 21.1 | 62.5          | 24.6 | 59.4 | 23.5 | 56.5 | 22.2 | 23.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 12.7          | 11.0 | 9.5 | DB                         | -17.0 | 30.7               | 3.0 | 1.8                                                | -19.1 | 31.9       | -20.9 | 33.0       | -22.5 | 34.0       | -24.7 | 35.3 | (4) |
| (5)               |      |     | WB                         | -17.4 | 22.7               | 2.9 | 0.6                                                | -19.5 | 23.1       | -21.2 | 23.5       | -22.8 | 23.9       | -24.9 | 24.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.3    | -2.6  | -2.4  | 0.3   | 5.4   | 10.1  | 14.9  | 19.5  | 19.7  | 16.5  | 11.4  | 5.9   | 0.7   |  |  |
|                                                                        | DBStd       | 8.68   | 4.96  | 4.43  | 4.11  | 3.18  | 2.87  | 2.98  | 2.37  | 2.07  | 2.97  | 3.49  | 4.03  | 4.63  |  |  |
|                                                                        | HDD10.0     | 1661   | 391   | 348   | 302   | 141   | 33    | 2     | 0     | 0     | 0     | 26    | 132   | 287   |  |  |
|                                                                        | HDD18.3     | 3769   | 649   | 581   | 559   | 387   | 256   | 108   | 14    | 9     | 70    | 217   | 372   | 546   |  |  |
|                                                                        | CDD10.0     | 1059   | 0     | 0     | 1     | 3     | 36    | 149   | 294   | 302   | 196   | 68    | 10    | 1     |  |  |
|                                                                        | CDD18.3     | 124    | 0     | 0     | 0     | 0     | 1     | 6     | 50    | 53    | 15    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 428    | 0     | 0     | 1     | 0     | 6     | 34    | 195   | 159   | 31    | 1     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 62     | 0     | 0     | 0     | 0     | 0     | 6     | 33    | 21    | 2     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 4.1    | 4.9   | 4.9   | 4.8   | 4.5   | 3.9   | 3.6   | 3.2   | 3.2   | 3.4   | 4.1   | 4.4   | 4.7   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1419   | 125   | 114   | 120   | 101   | 121   | 98    | 92    | 96    | 117   | 143   | 148   | 143   |  |  |
|                                                                        | PrecMax     | 1704   | 216   | 209   | 250   | 184   | 316   | 236   | 218   | 263   | 352   | 318   | 261   | 238   |  |  |
|                                                                        | PrecMin     | 1147   | 46    | 33    | 42    | 36    | 54    | 25    | 19    | 42    | 37    | 40    | 82    | 65    |  |  |
|                                                                        | PrecStd     | 155    | 39    | 47    | 54    | 35    | 56    | 49    | 45    | 52    | 77    | 62    | 49    | 44    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 10.7  | 10.1  | 14.3  | 17.1  | 23.6  | 26.9  | 29.8  | 29.3  | 26.2  | 22.3  | 16.5  | 13.3  |  |  |
|                                                                        |             | MCWB   | 10.2  | 8.8   | 5.8   | 9.9   | 15.2  | 19.2  | 20.5  | 20.9  | 19.4  | 16.9  | 14.8  | 12.5  |  |  |
|                                                                        | 2%          | DB     | 8.4   | 7.2   | 10.0  | 14.2  | 19.3  | 23.7  | 27.3  | 26.5  | 24.1  | 19.6  | 15.0  | 10.9  |  |  |
|                                                                        |             | MCWB   | 7.5   | 6.1   | 6.3   | 8.6   | 12.7  | 16.8  | 19.5  | 19.6  | 18.4  | 15.8  | 13.6  | 10.2  |  |  |
|                                                                        | 5%          | DB     | 6.6   | 5.4   | 7.6   | 12.3  | 16.9  | 21.8  | 25.4  | 25.0  | 22.4  | 17.9  | 13.7  | 9.5   |  |  |
|                                                                        |             | MCWB   | 5.7   | 4.2   | 4.8   | 8.3   | 11.8  | 15.9  | 19.1  | 18.8  | 17.6  | 15.2  | 12.5  | 8.7   |  |  |
|                                                                        | 10%         | DB     | 4.6   | 3.8   | 5.9   | 10.5  | 15.0  | 20.1  | 23.9  | 23.6  | 21.0  | 16.7  | 12.3  | 7.7   |  |  |
|                                                                        |             | MCWB   | 3.5   | 2.6   | 3.7   | 7.0   | 10.6  | 15.0  | 18.7  | 18.5  | 17.4  | 14.3  | 11.1  | 6.8   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 10.2  | 9.4   | 8.7   | 12.8  | 16.9  | 20.0  | 22.5  | 22.7  | 21.2  | 19.2  | 15.8  | 12.5  |  |  |
|                                                                        |             | MCDB   | 10.5  | 9.9   | 9.6   | 14.5  | 21.8  | 24.4  | 25.2  | 25.6  | 23.6  | 20.4  | 16.2  | 12.9  |  |  |
|                                                                        | 2%          | WB     | 7.8   | 6.6   | 6.7   | 10.7  | 14.5  | 18.6  | 21.5  | 21.6  | 20.1  | 17.0  | 14.1  | 10.2  |  |  |
|                                                                        |             | MCDB   | 8.2   | 7.3   | 7.9   | 12.7  | 17.3  | 22.0  | 24.6  | 24.6  | 22.0  | 18.1  | 14.7  | 10.8  |  |  |
|                                                                        | 5%          | WB     | 5.7   | 4.7   | 5.2   | 9.1   | 12.9  | 17.5  | 20.6  | 20.7  | 19.3  | 15.9  | 12.8  | 8.7   |  |  |
|                                                                        |             | MCDB   | 6.4   | 5.3   | 6.6   | 10.7  | 15.4  | 19.8  | 23.8  | 23.1  | 21.2  | 17.1  | 13.4  | 9.3   |  |  |
|                                                                        | 10%         | WB     | 3.5   | 2.8   | 3.6   | 7.7   | 11.4  | 16.3  | 19.7  | 19.8  | 18.3  | 14.7  | 11.1  | 6.8   |  |  |
|                                                                        |             | MCDB   | 4.4   | 3.7   | 5.3   | 9.8   | 13.9  | 18.8  | 22.5  | 21.9  | 20.4  | 16.1  | 12.0  | 7.5   |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 7.5   | 7.2   | 7.3   | 7.4   | 7.7   | 7.7   | 7.5   | 7.6   | 7.9   | 7.0   | 6.5   | 6.7   |  |  |
|                                                                        | 5% DB       | MCDBR  | 9.5   | 8.3   | 9.8   | 10.6  | 11.6  | 11.1  | 10.2  | 9.9   | 9.1   | 8.1   | 7.7   | 8.4   |  |  |
|                                                                        |             | MCWBR  | 9.8   | 8.0   | 7.3   | 6.8   | 6.4   | 5.6   | 4.3   | 4.2   | 5.1   | 5.2   | 7.2   | 8.9   |  |  |
|                                                                        | 5% WB       | MCDBR  | 9.6   | 8.6   | 8.6   | 8.6   | 9.4   | 8.9   | 8.0   | 7.4   | 7.2   | 6.4   | 7.2   | 8.3   |  |  |
|                                                                        |             | MCWBR  | 10.0  | 8.7   | 8.0   | 6.7   | 6.4   | 5.4   | 4.3   | 3.9   | 5.0   | 5.7   | 7.5   | 8.9   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.305 | 0.296 | 0.325 | 0.350 | 0.371 | 0.400 | 0.416 | 0.398 | 0.355 | 0.332 | 0.316 | 0.297 |  |  |
|                                                                        | taud        |        | 2.457 | 2.517 | 2.469 | 2.442 | 2.428 | 2.371 | 2.346 | 2.394 | 2.487 | 2.550 | 2.549 | 2.532 |  |  |
|                                                                        | Ebn at Noon |        | 816   | 902   | 913   | 913   | 899   | 870   | 850   | 854   | 869   | 846   | 802   | 795   |  |  |
|                                                                        | Edh at Noon |        | 71    | 80    | 96    | 107   | 112   | 119   | 121   | 111   | 93    | 77    | 65    | 60    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.43  | 2.31  | 3.51  | 4.29  | 4.94  | 5.38  | 5.37  | 4.99  | 4.05  | 2.62  | 1.50  | 1.12  |  |  |
|                                                                        | RadStd      |        | 0.11  | 0.17  | 0.24  | 0.36  | 0.63  | 0.50  | 0.39  | 0.35  | 0.27  | 0.27  | 0.16  | 0.09  |  |  |

## Historical Trends

|      |                         | DBAvg |     | Heating |        | Cooling |       | Degree-Days |         |         |         |         |
|------|-------------------------|-------|-----|---------|--------|---------|-------|-------------|---------|---------|---------|---------|
|      |                         |       |     | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP       | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e) | (f)     | (g)    | (h)     | (i)   | (j)         | (k)     | (l)     | (m)     |         |
| (46) | Station Only            | N/A   | N/A | N/A     | N/A    | N/A     | N/A   | N/A         | N/A     | N/A     | N/A     | (46)    |
| (47) | Regional (33 neighbors) | N/A   | N/A | N/A     | N/A    | +0.36   | +0.50 | N/A         | N/A     | +50     | +14     | (47)    |

Nomenclature: See separate page